

Quiz (Carolina Reaper)

Q1. Which of the following is a function?

- (A) $y = 2x$
- (B) $x = 2y$
- (C) $x^2 + y^2 = 4$
- (D) $y = \pm x$
- (E) $x = \pm y$

Q2. What is the domain of the function $f(x) = \frac{1}{x}$?

- (A) $(-\infty, \infty)$
- (B) $(-\infty, 0) \cup (0, \infty)$
- (C) $[0, \infty)$
- (D) $(-\infty, 0]$
- (E) $[0, \infty)$

Q3. Which of the following relations is one-to-one?

- (A) $y = x^2$
- (B) $y = 2x$
- (C) $x = 2y$
- (D) $x^2 + y^2 = 4$
- (E) $y = \pm x$

Q4. What is the range of the function $f(x) = x^2$?

- (A) $[0, \infty)$
- (B) $(-\infty, 0]$
- (C) $(-\infty, \infty)$
- (D) $[0, \infty)$
- (E) $(-\infty, 0)$

Q5. If $f(x) = 2x$ and $g(x) = x + 1$, what is $f(g(x))$?

- (A) $2x + 1$
- (B) $2x - 1$
- (C) $x + 2$
- (D) $x - 2$
- (E) $2x^2$

Q6. Which of the following statements is true about the function $f(x) = \frac{1}{x}$?

- (A) It is one-to-one and onto.
- (B) It is one-to-one but not onto.
- (C) It is onto but not one-to-one.
- (D) It is neither one-to-one nor onto.
- (E) It is a constant function.

Q7. What is the domain of the function $f(x) = \sqrt{x}$?

- (A) $[0, \infty)$
- (B) $(-\infty, 0]$
- (C) $(-\infty, \infty)$
- (D) $[0, \infty)$
- (E) $(-\infty, 0)$

Q8. If $f(x) = x^2$ and $g(x) = \sqrt{x}$, what is $f(g(x))$?

- (A) x
- (B) x^2
- (C) x^4
- (D) $\sqrt{x}$
- (E) x^3

Open Ended Questions (FRQ)**Q9.** Determine the domain and range of the function $f(x) = \frac{1}{x+1}$ and explain why it is a function.**Q10.** Given the relation $y = \pm\sqrt{x}$, explain why it is not a function and provide a counterexample.**Chef's Tips**

- Remind students to use proper function notation.
- Encourage students to draw graphs to visualize the functions.
- Emphasize the importance of understanding the definitions of domain, range, and function.